re LASL RFP 1956

The Los Alamos Scientific Laboratory wishes to obtain a large computer of advanced design to be available not later than the middle of 1959. The machine is to be used for general purpose scientific computing. One of the typical problems being, for example, hydrodynamic calculations in two dimensions requiring the random access to about 50,000 words of information, and computing speeds about a hundred times faster than those now available. The budget established for this purpose is limited to 3.5 million dollars.

The detail of the logics and order system are matters we would not wish to specify at this time but would rather work out with the group preparing the machine. The machine should, of course, be as advanced as the alloted time and funds permit and every effort should be made to incorporate important logical and electronic developments occurring during the period.

The main features should be of the type indicated below:

a) Orders should be performed in about one-half microsecond exclusive of access time, with the possible exception of multiplication and division which could be between one and two microseconds.

b) The machine must be capable of both fixed and floating point operations for words having about 15 decimal digits (or 50 binary digits).

c) The internal memory should be of the order of 50,000 words with a random access time of not over two microseconds per word, a buffer register being associated with each memory section.

d) A memory of smaller capacity for address arithmetic and for fast loops in the code with an access time appreciably shorter than the main memory may be desireable.

e) A certain amount of parallelism in control and address modification may be desireable to maximize the efficiency of the main arithmetic unit.

f) The uninterrupted error-free calculation sequence must be sufficiently long to permit good realization of the potential capability of the machine. This may require self-checking and self-correcting features in the design.

g) A fast buffered printer which should handle about 2,000 characters per second with automatic checking.

h) External memory equipment of large capacity capable of communicating with the internal memory at about two microseconds per digit, without essentially interrupting calculation.

i) The machine logic should be adaptable to automatic programming.